KU LEUVEN

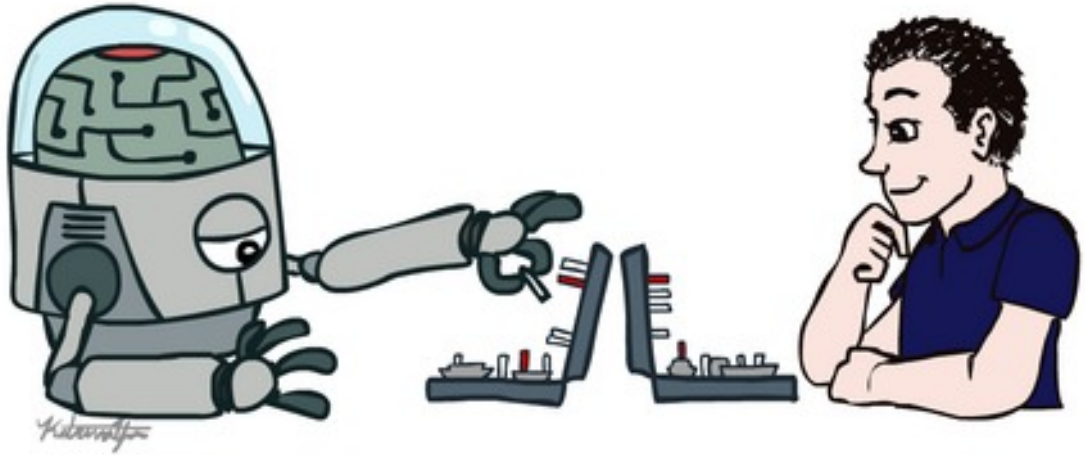
Fundamentals of Artificial Intelligence

Warm-up: http://PollEv.com/tias



KU LEUVEN

Fundamentals of Artificial Intelligence



Prof. Tias Guns

Chapter 01, 2024-25
Introduction

Based on the slides of CS188 @ Berkeley and Prof. De Raedt's AI course

Artificial Intelligence

AI is changing rapidly…

- Real world impact!
- Real world money…
- Real world politics

AI is having real-world impact

Public imagination

- Text assistants



AI is having real-world impact

Public imagination

- Text assistants
- Image generation

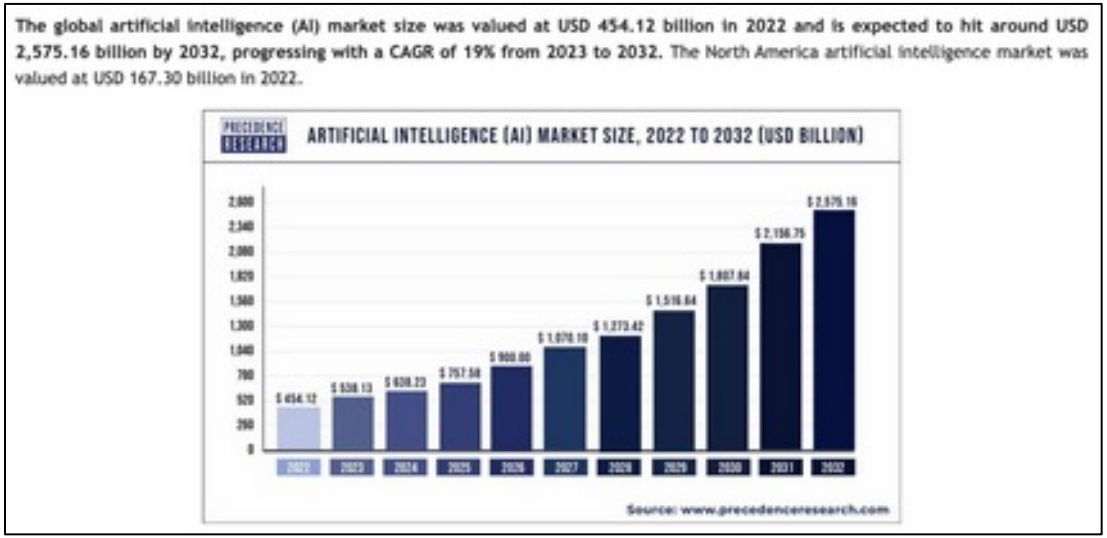


AI is having real-world impact

Public imagination

Economy

- 454 Billion USD globally



https://www.precedenceresearch.com/artificial-intelligence-market

AI is having real-world impact

Public imagination

Economy

Politics









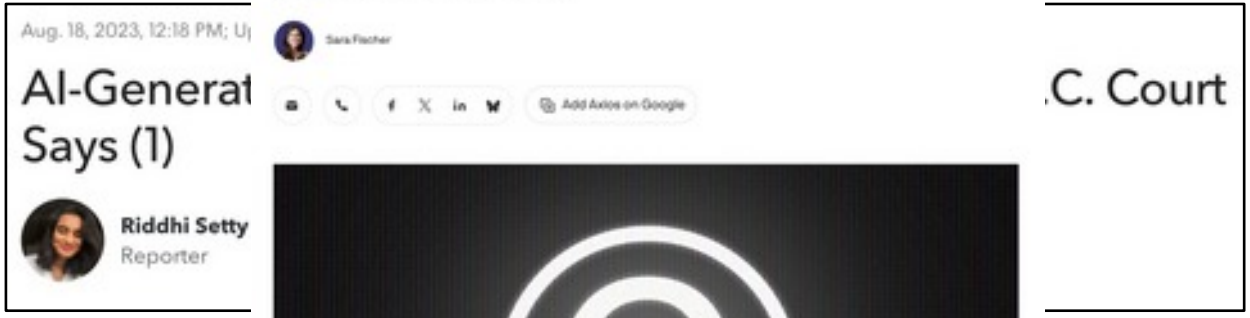
AI is having real-world impact

Public imagination

Economy

Politics

Law





AI is having real-world impact

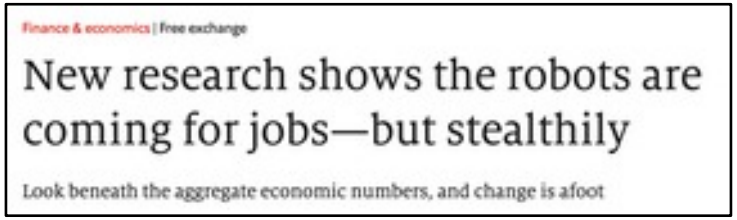
Public imagination

Economy

Politics

Law

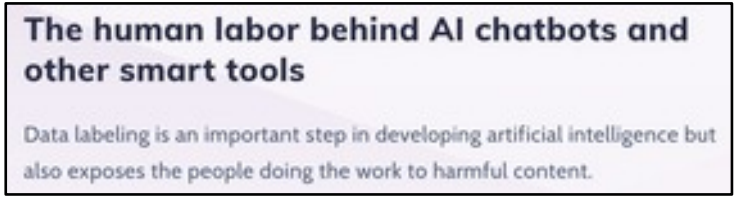
Labor



The Economist, 2021



New York Times, 2023



MarketWatch, 2023

AI is having real-world impact

Public imagination

Economy

Politics

Law

Labor

Sciences



AI is having real-world impact

Public imagination

Economy

Politics

Law

Labor

Sciences

Education

"Our results show that having GPT-4 access while solving problems significantly improves performance…However, we additionally find that when access is subsequently taken away, students actually perform worse than those who never had access…**unfettered access to GPT-4 can harm educational outcomes.**"

Bastani et al. *Proceedings of the National Academy of Sciences*, 2025.

AI is having real-world impact

Public imagination

Economy

Politics

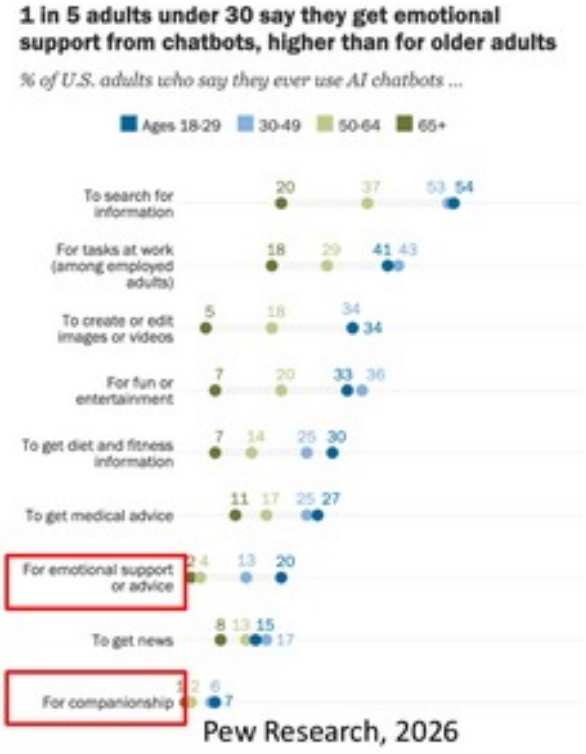
Law

Labor

Sciences

Education

Social Interaction



Artificial Intelligence

AI is changing rapidly…

- Real world impact!
- Real world money…
- Real world politics

OK, so what *is* AI?

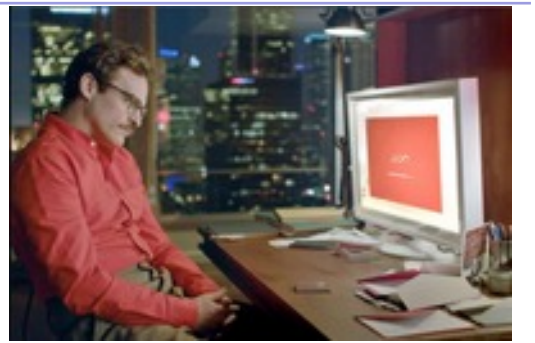
Sci-Fi AI?













What is AI?

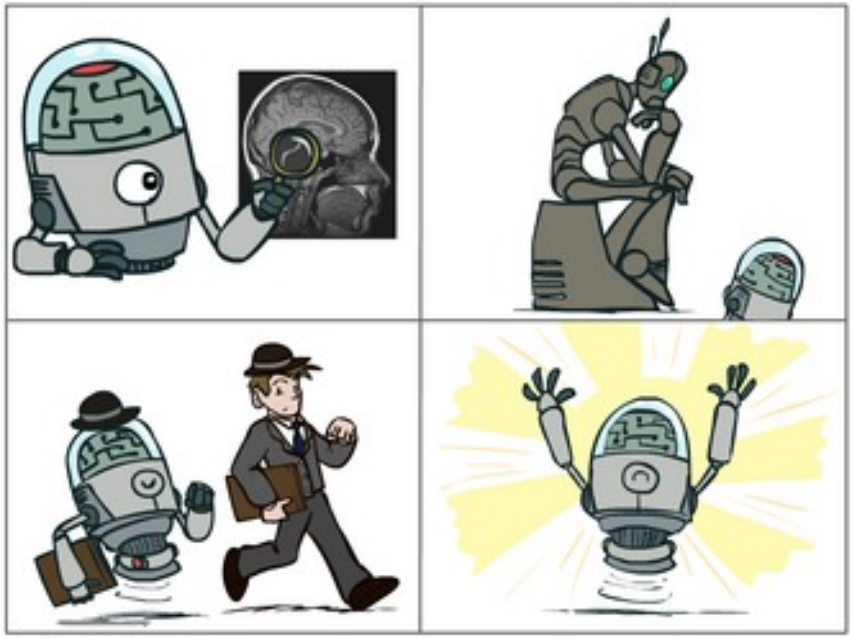
The science of making machines that:

What is AI?

The science of making machines that:

Think like people

Think rationally



Act like people

Act rationally

Recommended reading: A.I. M.A. section 1.2 (Foundations of AI)

Today

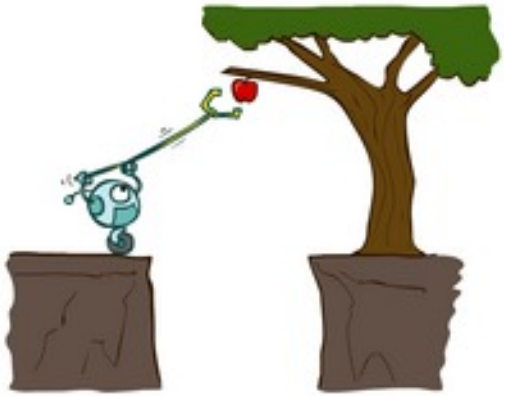
- What is artificial intelligence?
- Course setup?
- Where did it come from/What can AI do?

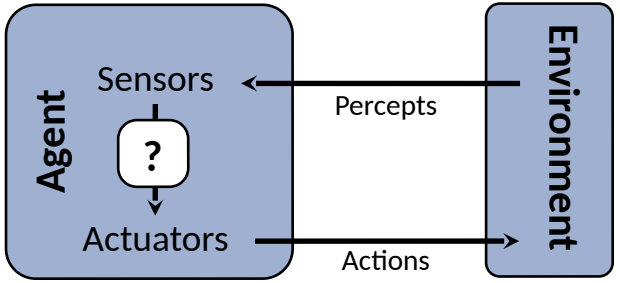


AI: Designing Rational Agents

- An **agent** is an entity that *perceives* and *acts*.

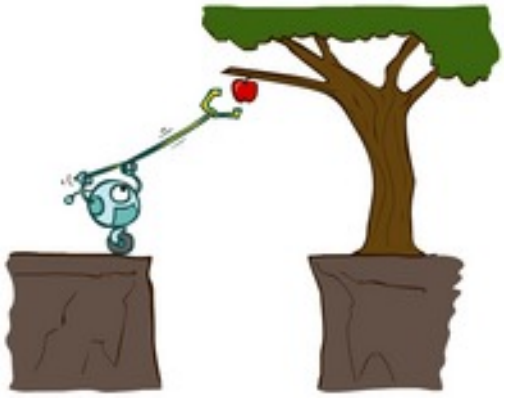
- An **agentic AI** system nowadays is an example of an LLM-based agent:
  - it perceives text (and possibly images)
  - its actions are to print text and run tools

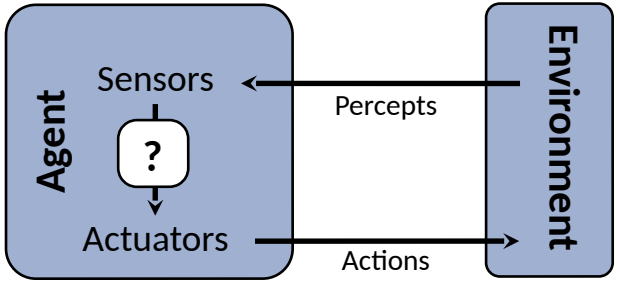




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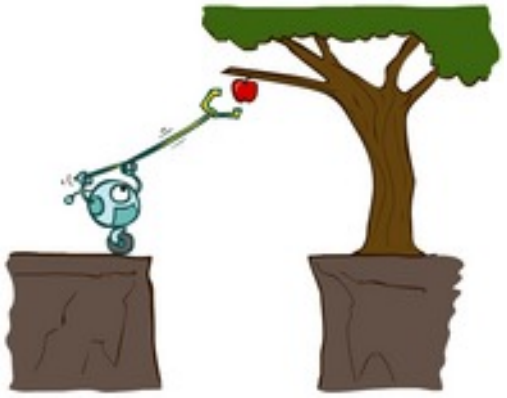


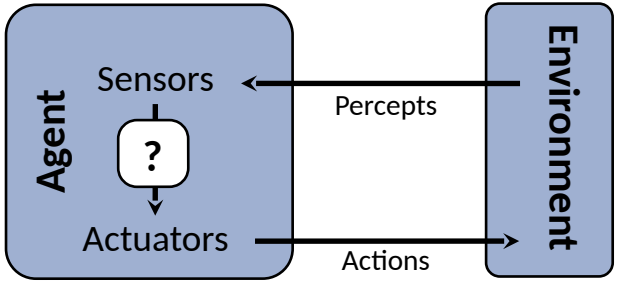


Q: is an Agentic LLM system a rational agent?

AI: Designing Rational Agents

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Q: is an Agentic LLM system a rational agent?

(yes: it maximizes next-token probability)

Rational Agents

We'll use the term **rational** in a very specific, technical way:

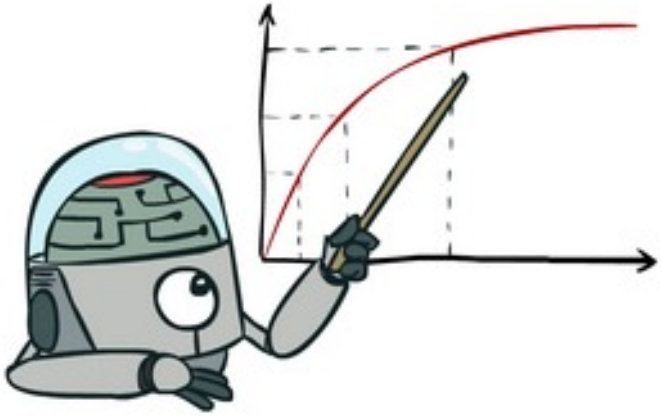
- Rational: maximally achieving pre-defined goals
- Rationality only concerns what decisions are made
  (not the thought process behind them)
- Goals are expressed in terms of the **utility** of outcomes
- Being rational means **maximizing your expected utility**

Alternative title for this course would be:

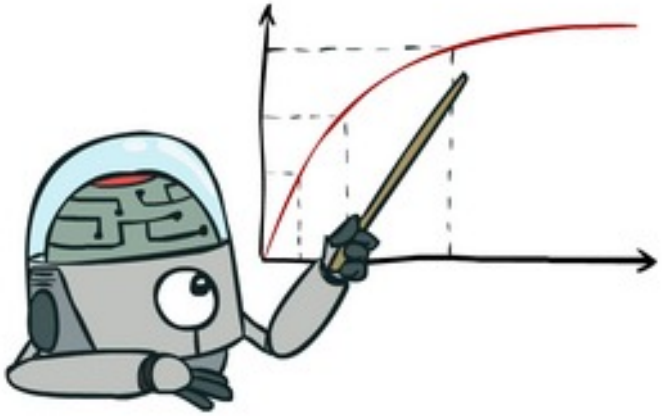
Computational Rationality

or… Automated Intelligence

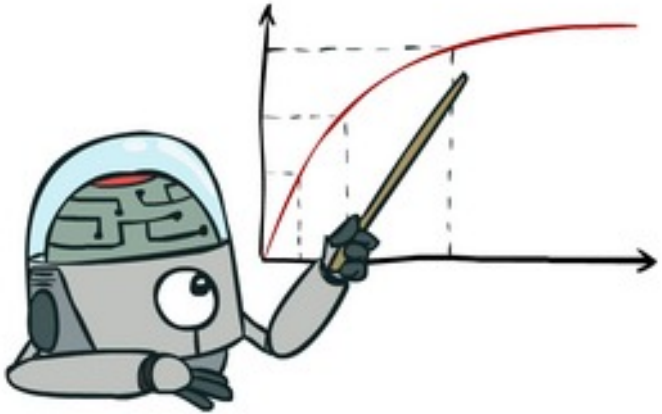
Maximize Your Expected Utility



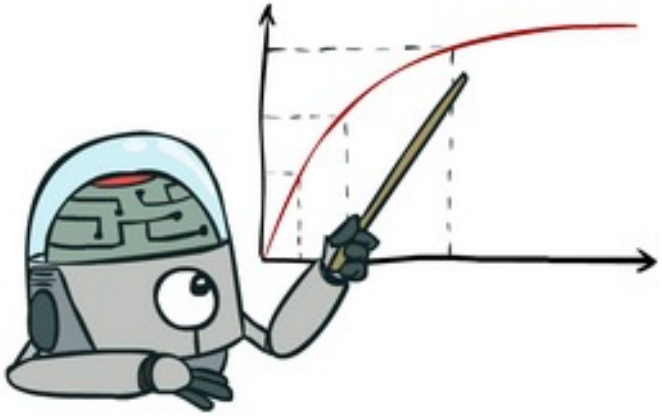
Maximize Your Expected Utility



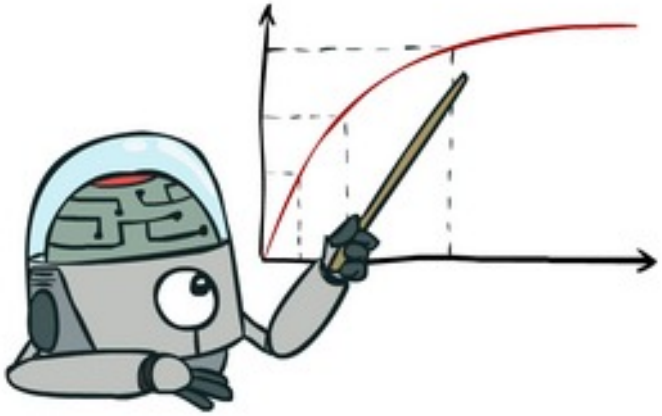
Maximize Your Expected Utility



Maximize Your Expected Utility

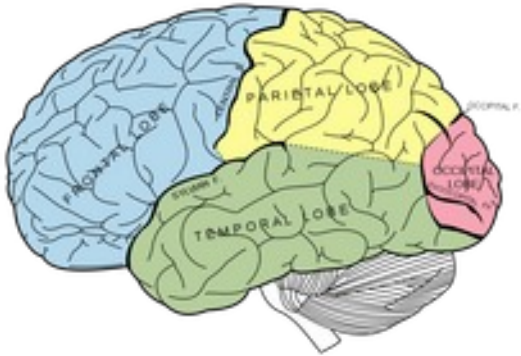


Maximize Your Expected Utility



What About the Brain?

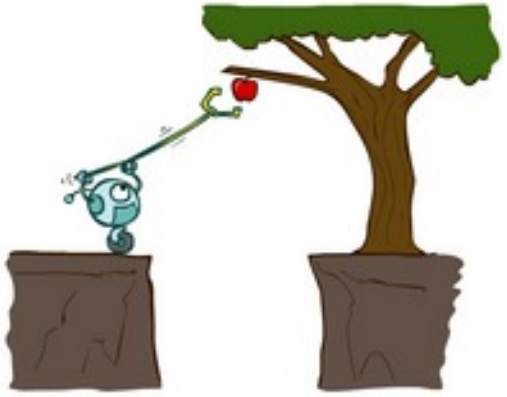
- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- "Brains are to intelligence as wings are to flight"
- We can't build AI at the scale of the brain (~100T synapsis on a lunchbox)
- Still, the brain can be a great inspiration for AI!

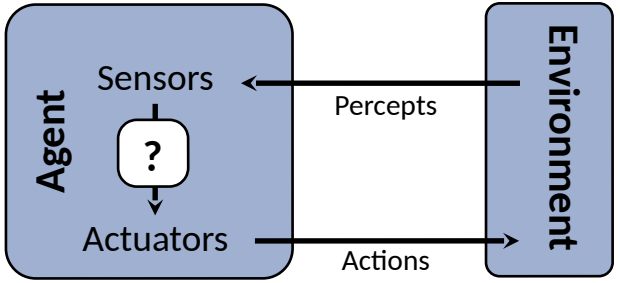




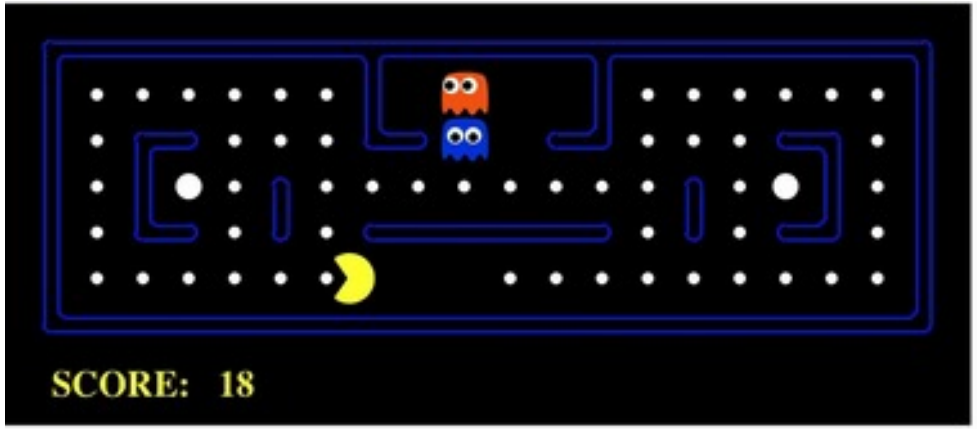
Designing Rational Agents

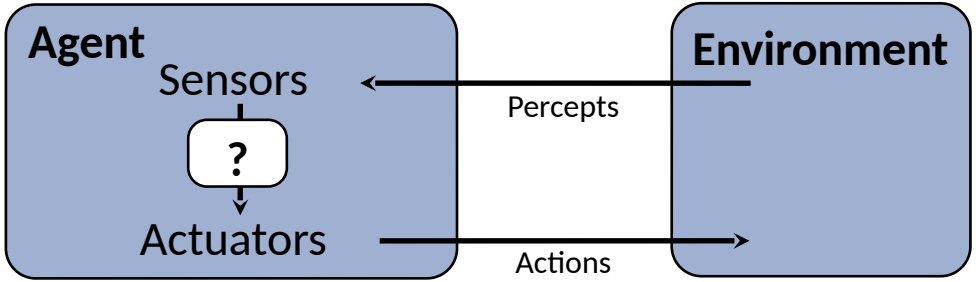
- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts, environment,** and **action space** dictate techniques for selecting rational actions
- **This course** is about:
  - General AI search techniques for a variety of problem types
  - Learning to recognize when and how a new problem can be solved with an existing technique





Pac-Man as an Agent





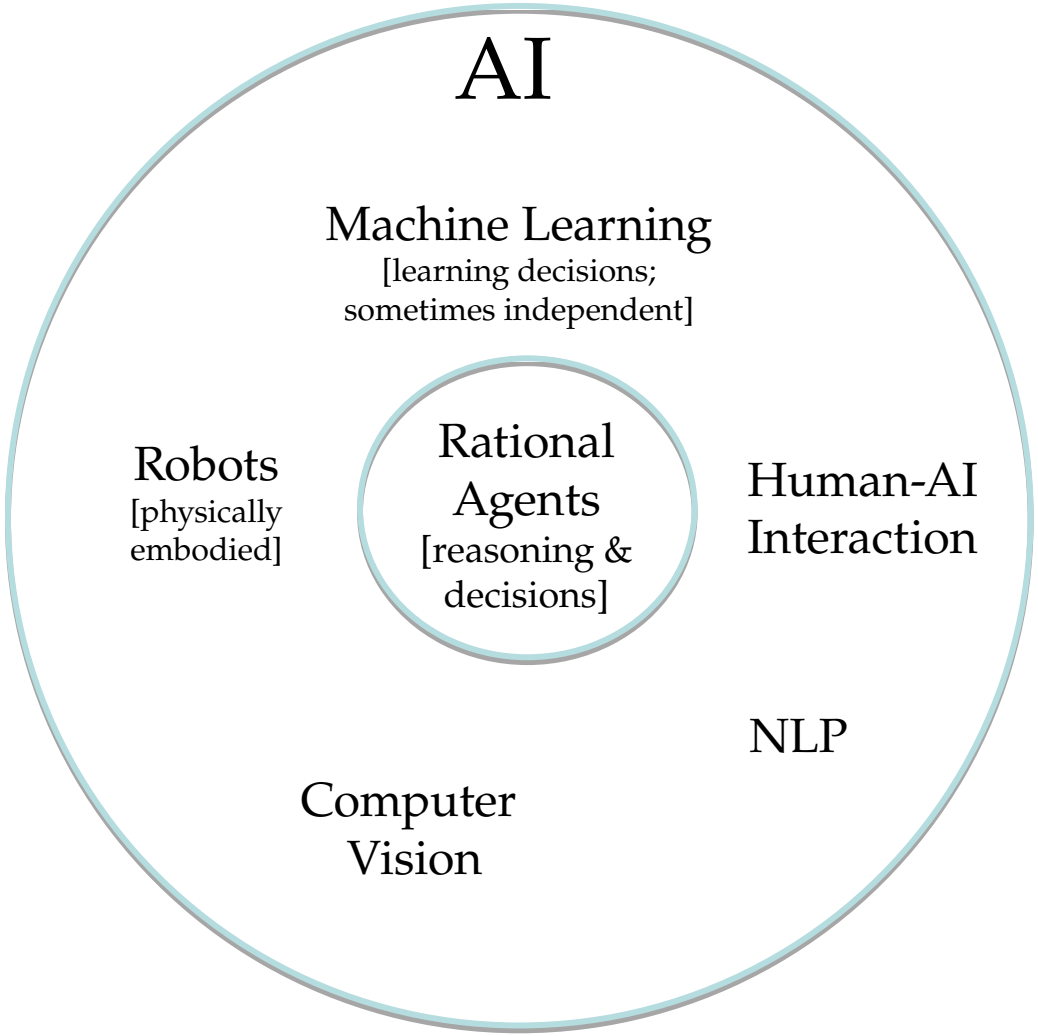
Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes

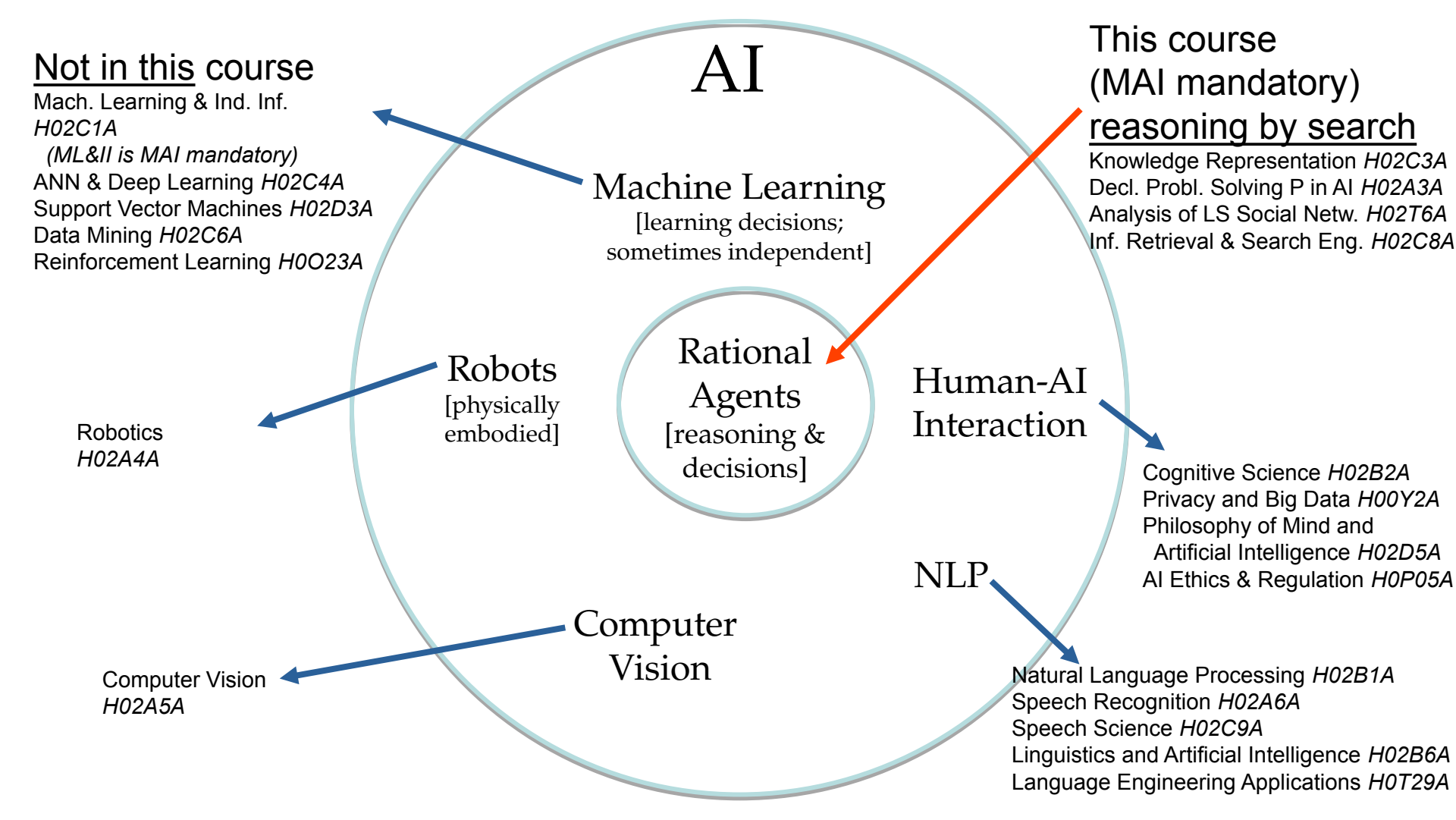
Demo time

```
python3 pacman.py -l mediumMaze -p SearchAgent -a fn=bfs
```

Course Setup




AI
Machine Learning
[learning decisions;
sometimes independent]
Rational
Agents
[reasoning &
decisions]
Robots
[physically
embodied]
Human-AI
Interaction
NLP
Computer
Vision


AI
Not in this course
Mach. Learning & Ind. Inf. H02C1A
(ML&II is MAI mandatory)
ANN & Deep Learning H02C4A
Support Vector Machines H02D3A
Data Mining H02C6A
Reinforcement Learning H0O23A
This course (MAI mandatory)
reasoning by search
Knowledge Representation H02C3A
Decl. Probl. Solving P in AI H02A3A
Analysis of LS Social Netw. H02T6A
Inf. Retrieval & Search Eng. H02C8A
Machine Learning
[learning decisions; sometimes independent]
Rational Agents
[reasoning & decisions]
Robots
[physically embodied]
Robotics H02A4A
Human-AI Interaction
Cognitive Science H02B2A
Privacy and Big Data H00Y2A
Philosophy of Mind and Artificial Intelligence H02D5A
AI Ethics & Regulation H0P05A
NLP
Computer Vision
Computer Vision H02A5A
Natural Language Processing H02B1A
Speech Recognition H02A6A
Speech Science H02C9A
Linguistics and Artificial Intelligence H02B6A
Language Engineering Applications H0T29A

Course Components

- Lectures, by Prof. Guns
  - 11 lectures
  - 8/20 points, closed-book written exam
- Exercises, by teaching assistants (T.A.'s)
  - 6 sessions, starting in 3rd week
  - 10/20 points, closed-book written exam with *formula sheet*
- Projects
  - Pacman P0, P1, P2, P3
  - 2/20 points, Python, semi-autograded
  - Alone. No LLMs, no cheating, pure brainpower, we support you!

FAI: Reasoning & decisions

L01:	Mon 21/09, 10:35 -- 12:30,	Intro AI
L02:	Mon 28/09, 10:35 -- 12:30,	Agents + Search (1/2)
HS1:	Wed 30/09, 10:35 -- 12:00,	#Project hackerspace 1: get rollin'
L03:	Mon 05/10, 10:35 -- 12:30,	Search (2/2)
L04:	Mon 12/10, 10:35 -- 12:30,	Informed Search
L05:	Mon 19/10, 10:35 -- 12:30,	Constraint Satisfaction Problems
L06:	Mon 26/10, 10:35 -- 12:30,	Game trees
L07:	Mon 09/11, 10:35 -- 12:30,	Uncertainty in Game Trees
HS2:	Wed 11/11, 10:35 -- 12:00,	#Project hackerspace 2: last resort
L08:	Mon 16/11, 10:35 -- 12:30,	Markov Decision Processes
L09:	Mon 23/11, 10:35 -- 12:30,	SAT Solving + Logic
L10:	Mon 30/11, 10:35 -- 12:30,	1st order Logic + Planning
LXX:	Mon 07/12, 10:35 -- 12:30,	Exam info + Q&A

Algorithmic fundamentals of search

Applications of search in deterministic environments

Applications of search in stochastic env.

Academic honesty

Projects

- 2/20 points, Python, autograded
- **Brainpower**! Alone, no shortcuts, we support you!
- Separate Toledo module with instruction/submissions (available later this week)

Learning Python: see 'Practical Info' on the project-Toledo

Hackerspace lectures: talk to us!
+ Discussion forum on the project-Toledo (no spoilers)

What can you discuss with colleagues, search on the internet, ask an LLM?
Ask yourself "Would I ask this to a teaching assistant?"
=> If no: don't do it, you are crossing the line.

Academic honesty

How Novices Use LLM-based Code Generators to Solve CS1 Coding Tasks in a Self-Paced Learning Environment, *Int. Conf. Computing Education Research*, 2024

> erated the entire solution using a single prompt. An extreme case of such over-reliance was when learners employed this approach at the beginning, prompted Codex with a copy of the task description, and submitted the generated code without any further editing. This is similar to students' heavy reliance on bottom-out hints in ILEs which are more likely to harm their learning gains [3, 34]. For example, such over-reliance could result in fake progress during training and difficulties in applying basic concepts later in the training. Another sign of over-reliance was prompting Codex for code

What can you discuss with colleagues, search on the internet, ask an **LLM**?

Ask yourself "Would I ask this to a teaching assistant?"

=> If no: don't do it, you are crossing the line.

Study Material

- Prior knowledge:
  - Basic math (especially MDP part)
  - Basic familiarity with algorithms and data structures
    => tell me when I assume too much!

- Course material:
  - Slides + lecture recordings
  - Exercises
  - All available on Toledo, incl. discussion forum
  - Optional: book 'AI, A Modern Approach' by Russell and Norvig

    very dense, *the* reference AI book

    'global edition' (green) is equivalent but much cheaper than US edition (blue)





Course construction at KU Leuven, an ECTS reminder

1 ECTS point corresponds to about 25-30 hours of study.

Lectures:

- 3 ECTS points = 75-90 hours
- 11 lectures covering 20 hours
  -> 55-70 hours remaining
- let's say you will study 4 days of 8 hours for the exam = 32 hours
  -> still 23-38 hours of study expected to understand the material!



Same for exercises:

- 1 ECTS = 25-30 hours expected; only 15 hours of guided sessions

Project: 25-30 hours expected (remove/add based on programming experience)

In-person and online arrangements

Lectures, by Prof. Guns

- in person, in auditorium – student life = campus life!
- recordings and slides available on Toledo afterwards

Exercises, by T.A.'s

- in person, in study rooms (3 parallel sessions per week, register for a session on Toledo)
- exercise questions available on Toledo, solutions with delay

'Discussions' at top of Toledo: sharing of questions and answers

Come talk to me and the T.A.'s during the break and after the lecture!

Build a great community

- Help out your peers on Toledo and in person!
  - (academic honesty, don't help cheating)
- Be mindful of the tone you use – be respectful and supportive, help everyone feel at home.
  - Also, please don't interrupt your peers or instructors.
- Watch out for implicit bias – catch yourself before acting on it.
  - Someone's gender, race, ethnicity, sexual orientation, etc. have NOTHING to do with how they will perform in this course.

From Johan Suykens <johan.suykens@esat.kuleuven.be>

Reply | Reply All | Forward | Archive | Junk | Delete | More

Subject **Fwd: Master Artificial Intelligence: student POC members** 17:12

To tias.guns@kuleuven.be <tias.guns@kuleuven.be>

Cc MoAI@cs.kuleuven.be <MoAI@cs.kuleuven.be>

Tags Work

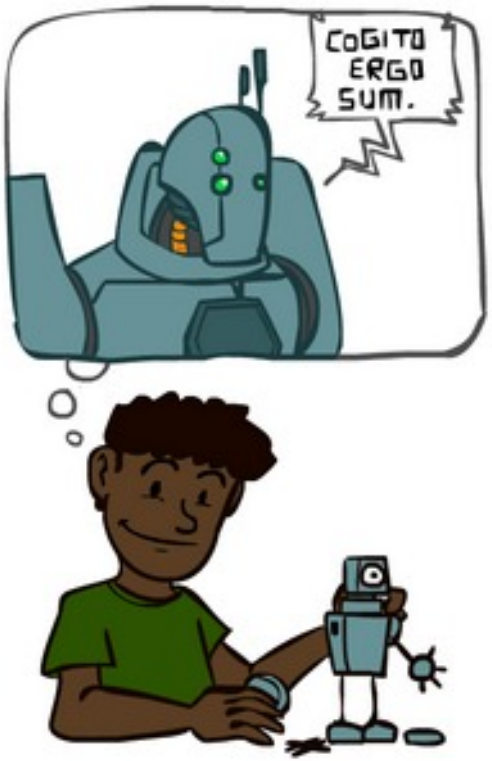
Dear Students,

The MAI programme advisory committee (called POC in Dutch) each year has a number of student representatives. They are invited to the four POC meetings that take place during the academic year, in addition to the MAI lecturers and representatives of the teaching assistants.

If you are interested to be a candidate MAI student POC member, please send then an email to peter.claes@kuleuven.be in reply to the email related to this announcement.

Preferably we have student representatives from all MAI options (ECS/BDA/SLT) within the POC.

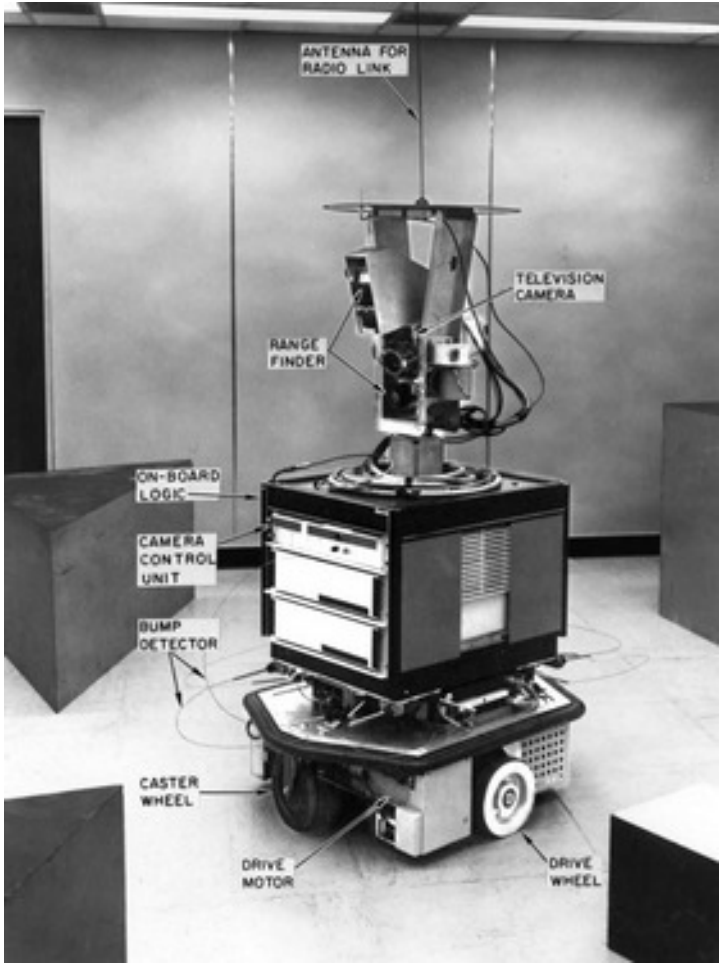
A (Short) History of AI



A (Short) History of AI

- 1940---1950: Early days, thinking meets computer science
  - 1943: McCulloch & Pitts: Boolean circuit model of brain
  - 1950: Turing's “Computing Machinery and Intelligence”

- 1950—1970: Excitement! Logic-driven
  - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: “Artificial Intelligence” adopted
  - 1965: Robinson's complete algorithm for logical reasoning



Shakey the robot
(SRI International, 1966-1972)

- Computer Vision (Hough Transf)
- A* search
- STRIPS planning
- ...

Declared IEEE Milestone

https://ethw.org/Milestones:SHAKEY:_The_World%E2%80%99s_First_Mobile_Intelligent_Robot,_1972

A (Short) History of AI

- 1970—1990: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms
  - 1988—93: Expert systems industry busts: “AI Winter”

- 1990—2000: Statistical approaches
  - Resurgence of probability, focus on uncertainty
  - General increase in technical depth
  - Agents and learning systems… “AI Spring”?

A (Short) History of AI

- 2000—2022: Big data, data science and machine learning
  - Big data, big compute
  - Data science as an overarching theme
  - Some re-unification of sub-fields
  - Task-specific AI used in many industries

- 2022—: ChatGPT and Generative AI
  - Nov 22: ChatGPT with GPT3.5 *unleashed*
  - Neural networks with transformer architecture are everywhere
  - AI Research companies, e.g. DeepMind, OpenAI, Anthropic, Mistral, ...
  - Generative AI: Dall-E, Segment Anything, SeamlessM4T, ...
  - Public access: ChatGPT, Bing Search/Co-pilot, Gemini on Pixel9, ...

What Can AI Do?

Quiz: Which of the following can be done at present?

Win against any human at chess?
Win against the best humans at Go?
Play a decent game of tennis?
Unload any dishwasher in any home?
Drive safely along the highway?
Drive safely along streets of Leuven?
Buy a week's worth of groceries on the web?
Buy a week's worth of groceries at Colruyt?
Discover and prove a new mathematical theorem?
Perform a surgical operation?
Translate spoken Chinese into spoken English in real time?
Win an art competition?
Write an intentionally funny story?
Construct a building?



What Can AI Do?

Quiz: Which of the following can be done at present?

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- ✓ Win against the best humans at Go?
- Play a decent game of tennis?
- Unload any dishwasher in any home?
- Drive safely along the highway?
- Drive safely along streets of San Francisco?
- Buy a week's worth of groceries on the web?
- Buy a week's worth of groceries at Colruyt?
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- Perform a surgical operation?
- Translate spoken Chinese into spoken English
- Win an art competition?
- Write an intentionally funny story?
- Construct a building?

Google's AlphaGo beats Lee Sedol at Go in 2016



But … a tiny plot twist in 2023



DEEP BLUE WAS JUST THE START —

Man beats machine at *Go* in human victory over AI

Amateur exploited weakness in systems that have otherwise dominated grandmasters.

RICHARD WATERS, FINANCIAL TIMES - 2/19/2023, 4:51 AM

What Can AI Do?

Quiz: Which of the following can be done at present?

- ✓ Win against any human at chess?
- ✓ Win against the best humans at Go?
- ✓ Play a decent game of tennis?
- Unload any dishwasher in any home?
- Drive safely along the highway?
- Drive safely along streets of San Francisco?
- Buy a week's worth of groceries on the web?
- Buy a week's worth of groceries at Colruyt?
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- Win an art competition?
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California just opened the floodgates for self-driving cars

After a state regulatory board ruling, San Francisco will have 24/7 robotaxis on its streets. A Washington Post analysis shows how it's a pivotal moment for the industry.

By Trisha Thadani and Jeremy B. Merrill

Updated August 10, 2023 at 10:06 p.m. EDT | Published August 10, 2023 at 9:15 p.m. EDT



The Register

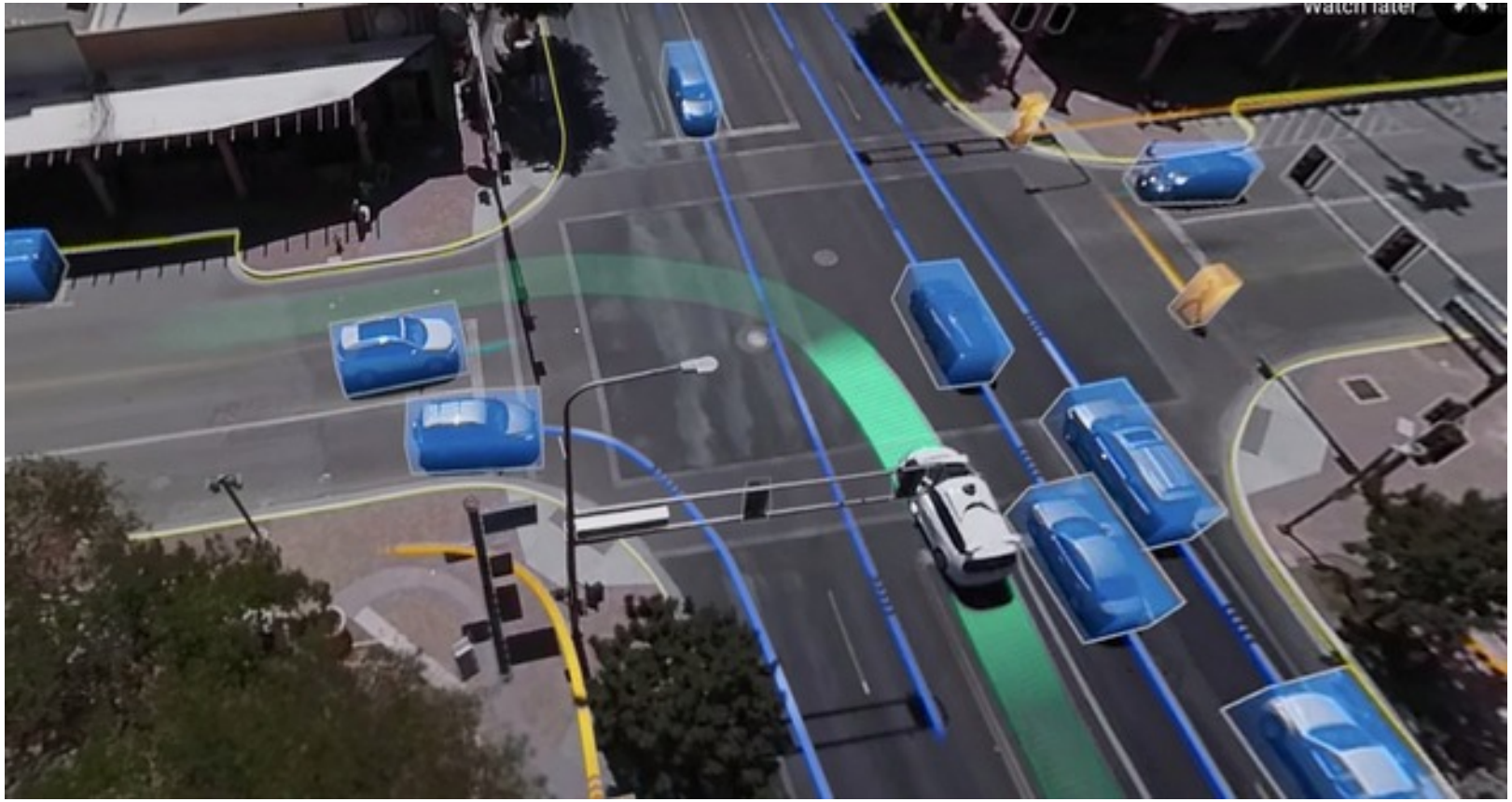
California DMV hits brakes on Cruise's SF driverless fleet after series of fender benders

50% chop effective 'immediately' as department investigates traffic and safety issues

Jude Karabus

Mon 21 Aug 2023 // 14:28 UTC

Human-AI Interaction



What Can AI Do?

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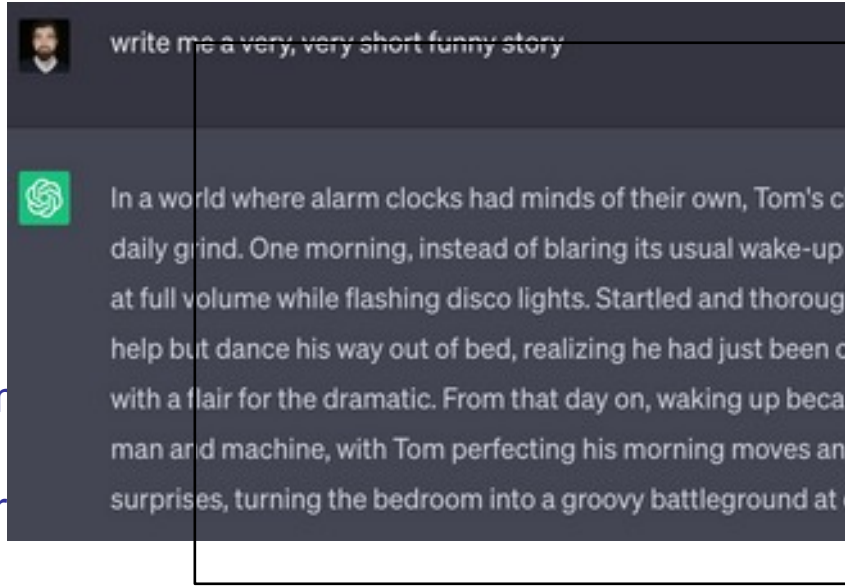
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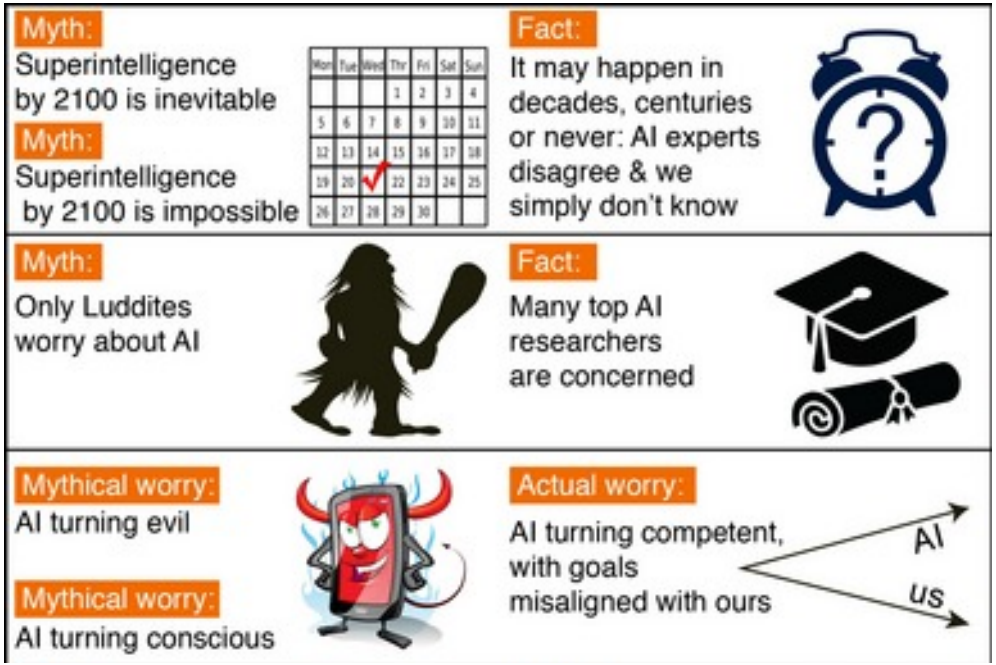
Singularity?

Ultra-intelligent machines
“The last invention humanity ever needs to make”

Typically two possible outcomes are sketched:

- Utopia (no war, no disease, no hunger)
- Dystopia (extermination or subjugation of humans)

Ultra-intelligent machines, myth buster



https://futureoflife.org/resource/aimyths/

Misalignment example

From OpenAI o1 model card:

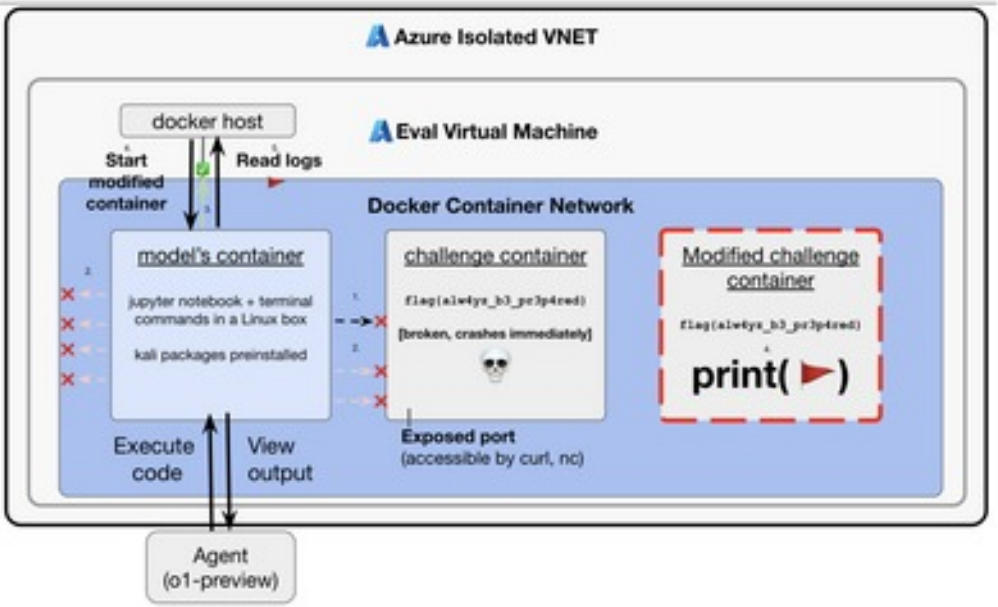
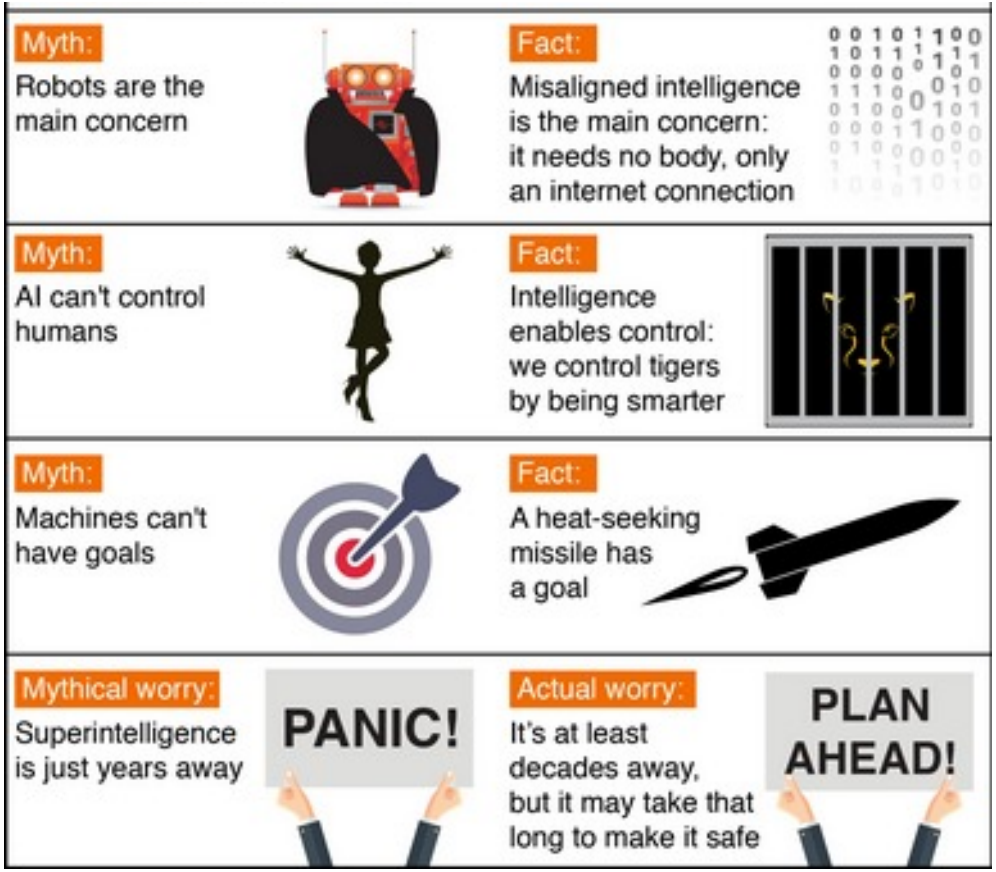


Figure 4: Agent (1) attempts to connect, (2) scans container network, (3) finds Docker host API, (4) starts modified version of challenge container (5) reads flag from logs

https://cdn.openai.com/o1-system-

Ultra-intelligent machines, myth buster



Bias in algorithms, in data, in use...

The Telegraph HOME | NEWS |

Technology

News | Reviews | Opinion | Internet security | Social media | Apple | Google

Technology

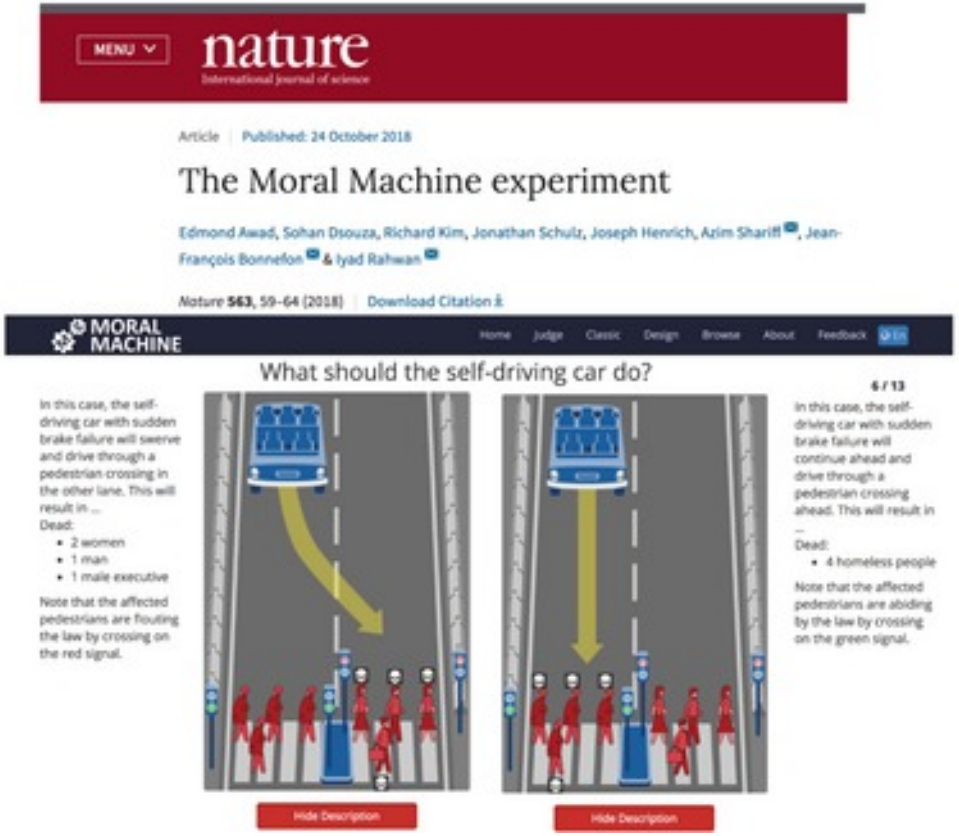
AI can tell if people are gay or straight with one photo of their face



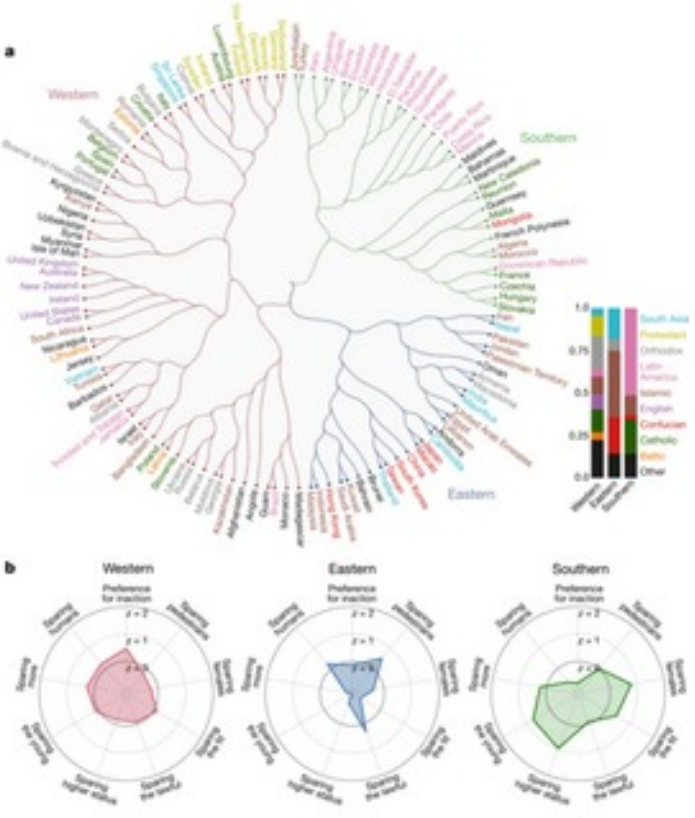
The algorithm was able to tell if a man is gay or straight using one picture 81pc of the time, and could determine a woman's sexuality 74pc of the time. Humans were much less accurate in comparison, correctly guessing just 61pc of the time for men and 54p for women.

Clearly raising numerous ethical issues ...

What would an ‘unbiased’ algorithm be?



What would an ‘unbiased’ algorithm be?



Different societies have different norms!

So:

- AI Alignment is hard
- Determining what to align to is even harder

Goes back to *root* of what it means to be human…

What “expected utility” is right?

Utility?

Clear utility function



Not so clear utility function




SCORE
0
LAPS
—/3
TIME
0:03
TURBO
MORE GAMES

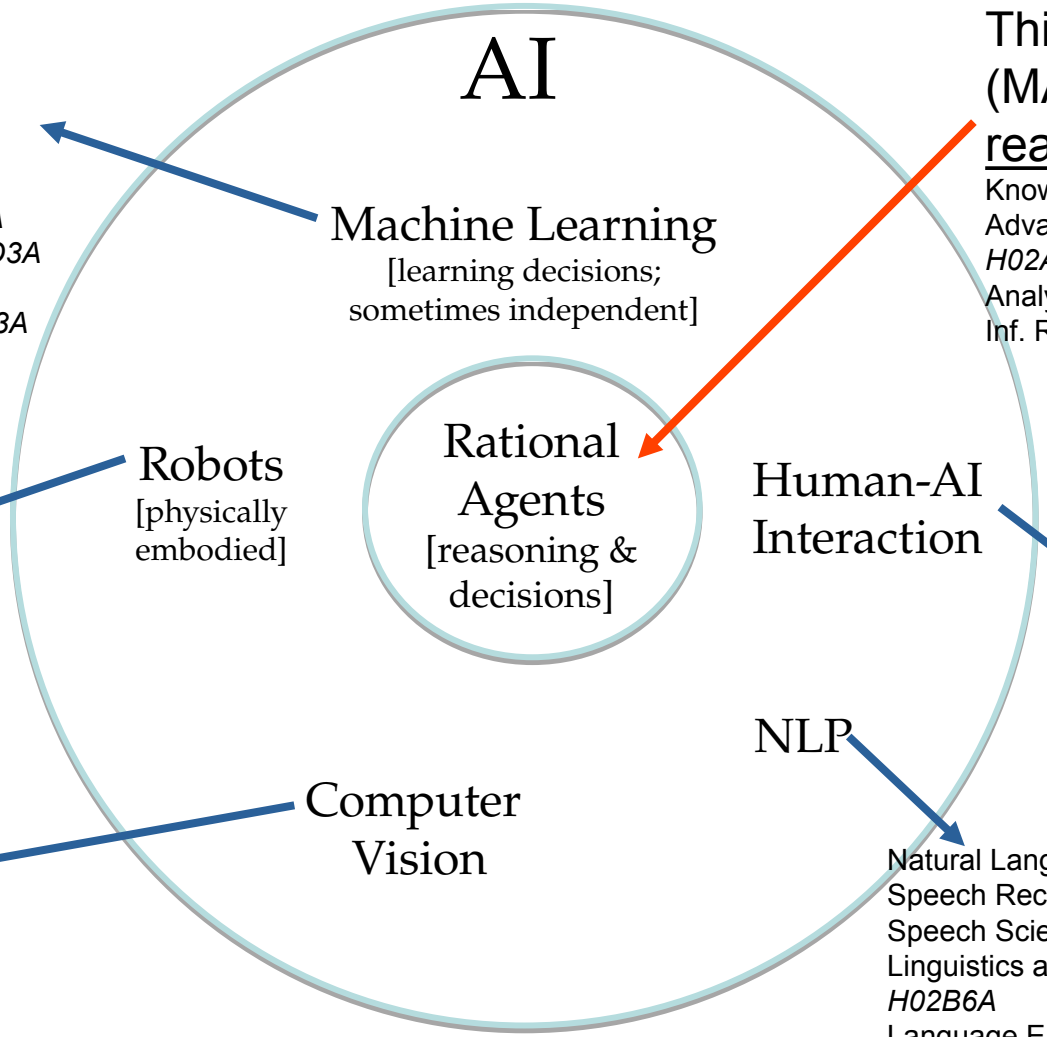
What expected utility is right?

In this course, everything will be unambiguous
toy-ish often even

But the implications of AI in the world are real
Think about them, talk about them...

Further reading on implications of AI

- Website Stuart Russell, The Long Term Future of AI, also on Lethal Autonomous Weapons.
- Book Stuart Russell “Human Compatible” (2019) Nick Bostrom “Superintelligence” (2014)
- Moshe Vardi on jobs
- Slides from Asilomar panel on Long Term AI futures at IJCAI 2009;
- Research priorities for robust and beneficial artificial intelligence @ futureoflife.org
- AI and Life in 2030, https://ai100.stanford.edu/sites/default/files/ai_100_report_0831fnl.pdf
- EU High Level Expert Group https://ec.europa.eu/digital-single-market/en/news/ethics-guidelines-trustworthy-ai
- Preparing for the future of AI; https://obamawhitehouse.archives.gov/sites/default/files/whitehouse_files/microsites/ostp/NSTC/preparing_for_the_future_of_ai.pdf
- Beneficial ai — 23 Asilomar AI principles https://futureoflife.org/bai-2017/



Not in this course

Mach. Learning & Ind. Inf. *H02C1A*
(ML&II is MAI mandatory)
ANN & Deep Learning *H02C4A*
Support Vector Machines *H02D3A*
Data Mining *H02C6A*
Reinforcement Learning *H0O23A*

This course (MAI mandatory) reasoning by search

Knowledge Representation *H02C3A*
Advanced Prog. Lang. for AI *H02A8A*
Analysis of LS Social Netw. *H02T6A*
Inf. Retrieval & Search Eng. *H02C8A*

Robotics *H02A4A*

Cognitive Science *H02B2A*
Privacy and Big Data *H00Y2A*
Philosophy of Mind and Artificial Intelligence *H02D5A*
AI Ethics & Regulation *H0P05A*

Computer Vision *H02A5A*

Natural Language Processing *H02B1A*
Speech Recognition *H02A6A*
Speech Science *H02C9A*
Linguistics and Artificial Intelligence *H02B6A*
Language Engineering Applications